MILITARY SPECIFICATION

FLASHLIGHT, EYE EXAMINING, DISPOSABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers disposable, self-contained, diagnostic flashlights used in ophthalmic procedures.

1.2 Classification. The flashlights shall be of the following types as specified (see 6.2):

Type I - 1 millimeter (mm) aperture, unfiltered.
Type II - Spot beam, cobalt blue filter.
Type III - Slit beam, cobalt blue filter.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corners.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Boxes, Setup.

MILITARY

MIL-P-116 - Preservation-packaging, Methods Of.
MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6515

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

C18.1 - Dry Cells and Batteries, Specifications for.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

THE AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E308 - Spectrophotometry and Description of Color in CIE 1931 System, Standard Recommended Practice for.

(Application for copies should be addressed to The American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 First article. When specified in the procurement document, the contractor shall furnish sample unit(s) for first article inspection and approval (4.1.2.1 and 6.2).

3.2 Material. Materials shall be of the highest quality throughout and as specified herein. Materials shall be suitable in all respects for their intended purpose and use.

3.2.1 Batteries. The batteries shall be 1.5 volt, alkaline batteries complying with battery designation L40 of ANSI C18.1.

3.2.2 Lamp. The lamp shall be of a type designed for use with two batteries as specified in 3.2.1. The glass envelope shall be of a design to meet the operational requirements for the various types of flashlights specified herein (see 6.6). The lamp for the type III flashlight shall be equipped with a straight wire filament to prevent the filament outline from being projected with the beam of light. The lamp for the type III flashlight shall have a design voltage of 2.25 to 2.33 volts and a design current of 0.25 to 0.27 amperes.

3.3 Design and construction. The flashlight shall be a disposable, self-contained, diagnostic flashlight used in ophthalmic procedures. The flashlight shall be 6-1/2 $\pm$ 1/4 inches long, 5/8 $\pm$ 1/16 inch in diameter, and have a weight of 2-1/2 $\pm$ 1/2 ounces. The flashlight shall consist of a housing, batteries, bulb, an on-off switch, a pocket clip and any additional components necessary for the proper functioning of the flashlight. The flashlight shall also include a filter, lens and/or mask as required by the specified type.

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3.3.1 Lens, mask and filter.

3.3.1.1 Type I. The type I flashlight shall be equipped with a plastic mask with a 1 ± 0.01 millimeter (mm) diameter aperture. The flashlight shall be capable of producing an intense point of light for use as a patient fixation source.

3.3.1.2 Type II. The type II flashlight shall be equipped with a cobalt blue filter for use in conjunction with fluorescein dye. When tested as specified in 4.4.3, the flashlight shall be capable of producing blue light having a dominant wave length of 453.6 to 501.4 millimicrons (mu) with an excitation purity of 87.1 to 96.3 percent with an illuminant having a color temperature of 2854 Kelvin (K). The luminous transmittance shall be 3.4 ± 0.17 percent (see 6.5). When tested as specified in 4.4.1 the flashlight shall be capable of producing a spot beam of cobalt blue light within a 4-1/2 inch diameter circular area.

3.3.1.3 Type III. The type III flashlight shall be equipped with a cobalt blue filter, for use in conjunction with fluorescein dye, and a slit mask. The flashlight shall be capable of producing a sharply defined slit beam, of cobalt blue light, 7/16 to 9/16 inch in length, and a maximum width of 1/16 inch, when tested as specified in 4.4.2. When tested as specified in ASTM E308 the flashlight shall be capable of producing blue light having the characteristics specified in 3.3.1.2.

3.3.2 Housing. The housing shall be fabricated of rigid thermoplastic tubing, crimped or formed at each end to secure the internal components. The tubing shall have a minimum wall thickness of 0.024 inches.

3.3.3 Switch. The switch shall be of the rotary type or slide type. If the rotary type switch is used, the switch knob shall be fabricated of plastic or other suitable material and shall be located at the end of the flashlight. If the slide switch type is used it shall be an integral part of the pocket clip (see 3.3.6).

3.3.4 Batteries. The batteries used in the assembly of the flashlight shall comply with the requirements of 3.2.1. The flashlight shall be supplied with two batteries.

3.3.5 Lamp. The lamp shall be as specified in 3.2.2. The lamp shall be selected to provide for minimum current drain.

3.3.6 Pocket clip. The flashlight shall be furnished with a polished metal pocket clip securely attached to the plastic tube at the switch end of the flashlight. The design of the pocket clip shall be such that there are no sharp or rough edges.

3.4 Operational. The flashlight shall illuminate immediately without any jiggling of the switch when tested as specified in 4.4.4.

3.5 Inspection/test date. The inspection/test date shall be 24 months after date of manufacture.

3.5.1 Delivery. Not more than 3 months shall have elapsed from the date of manufacture to the date of delivery to the Government.

3.6 Identification marking. Each flashlight shall be permanently and legibly marked with the following:

(a) Name or registered trademark of the manufacturer.
(b) Manufacturer's model number.
(c) Type of flashlight.
(d) National stock number.

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3.7 Workmanship. Workmanship shall be first class throughout. The flashlight shall be free from defects which detract from its appearance or may impair its serviceability. All burrs and sharp edges shall be removed from the flashlight.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection.
b. Quality conformance inspection.

4.1.2.1.1 First article inspection. First article inspection shall be performed on the flashlight when a first article sample is required (see 3.1). This inspection shall include examination of 4.3 and tests of 4.4.

4.1.2.1.2 Quality conformance inspection. Quality conformance inspection shall include the examination of 4.3 and tests of 4.4.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the materials may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

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4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table I. The unit of product for sampling purposes shall be one flashlight of each type.

TABLE I. Sampling for examination.

Examination	Inspection level	AQL (percent defective)
For visual examination		
Major defects	S-4	1.5
Minor defects	S-4	4.0
For dimensional examination	S-1	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table II. The unit of product for sampling purposes shall be one flashlight of each type.

TABLE II. Sampling for tests.

Component	Characteristic	Requirement	Test procedure	Inspection level	AQL (percent defective)
Filter	Color	3.3.1.2	4.4.3	S-1	4.0
		3.3.1.3	4.4.3	S-1	4.0
Flashlight	Spot beam	3.3.1.2	4.4.1	S-1	4.0
	Slit beam	3.3.1.3	4.4.2	S-1	4.0
	Operational	3.4	4.4.4	S-2	1.0

4.3 Examination.

4.3.1 Flashlight. The flashlight shall be examined for defects including, but not limited to, those listed in table III.

TABLE III. Classification of defects.

Category	Defect
Major:	
101	Flashlight not of correct type.
102	Pocket clip not securely fastened.
103	Switch does not operate smoothly.
104	Ends of tube not sufficiently rolled or crimped to prevent internal components from rattling.
105	Flashlight not complete with all components.
Minor:	
201	Identification marking not complete, correct, legible or permanent.
202	Case not free of scratches, rough edges, dirt, grease or foreign matter.
203	Dimensions not as specified.

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4.3.2 Packaging inspection. The inspection of the preservation-packaging and interior package marking shall be in accordance with the group A and B quality conformance inspection requirements of MIL-P-116. The inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with requirements specified herein.

4.4.1 Spot beam. The bulb end of the flashlight shall be supported perpendicular to, and 4 ± 1/4 inch above a horizontal surface. The spot shall have a maximum diameter of 4-1/2 inches, centered at a point where the projection of the bulb axis intersects the horizontal surface.

4.4.2 Slit beam. The bulb end of the flashlight shall be supported perpendicular to, and 1-3/8 ± 1/8 inches above a horizontal plane. The slit beam shall meet the requirements specified in 3.3.1.3.

4.4.3 Color. The transmission characteristics for the cobalt blue filter shall be as specified in 3.3.1.2 when tested in accordance with ASTM E308.

4.4.4 Operational test. The flashlight switch shall be actuated for three on-off cycles. The flashlight shall meet the requirements specified in 3.4.

5. PACKAGING

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Unit of issue. One flashlight of the type specified constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit packaging. Each flashlight shall be clean, dry and packaged in accordance with MIL-P-116, method III in a box conforming to PPP-B-566 or PPP-B-676. Closure shall be as specified in the appendix of the applicable box specification. As an alternate, each flashlight may be blister packaged.

5.1.3 Level C. Flashlights shall be clean, dry and individually packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A. One hundred flashlights of one type only shall be packed in a shipping container designed for a type 2 load and conforming to PPP-B-636, class weather-resistant. Closure, strapping and waterproofing shall be as specified in the appendix of the box specification.

5.2.2 Level B. One hundred flashlights of one type only shall be packed in a shipping container designed for a type 2 load and conforming to PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

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5.2.3 Level C. The packaged flashlights shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the carrier's rules and regulations.

5.3 Marking.

5.3.1 Unit package. Each unit package shall be marked in accordance with MIL-STD-129. Contract or purchase order number and name of contractor shall be shown. Type II shelf life marking as specified in MIL-STD-129 shall be shown. Marking shall include the legend:

"STORE BETWEEN 2° - 8° C (35° - 46° F)".

5.3.2 Shipping container. Shipping containers shall be marked as specified in MIL-STD-129. Type II shelf life marking as specified in MIL-STD-129 shall also be shown. Marking shall include the legend:

"STORE BETWEEN 2° - 8° C (35° - 46° F)".

NOTE: The unrefrigerated shipping time for this item shall be not more than thirty days (720 hours). DD Form 1502, 1502-1, or 1502-2 as specified in MIL-STD-129 shall not be required.

6. NOTES

6.1 Intended use. The pocket flashlights covered by this specification are intended for use in ophthalmic procedures such as detecting abrasions and defects of the cornea, diagnosing corneal injuries, fitting contact lenses and conducting various muscle tests of the human eye.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

a. Title, number and date of this specification.
b. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
c. Type and size required (see 1.2).
d. When a first article is required (see 3.1, 4.1.2.1 and 6.3).

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of the Armed Services Procurement Regulations. The contract or purchase order specifies the quantity, procedure for delivery and other specific instructions for inspection and approval of the first article.

6.4 This specification covers the following item(s) appearing in the Federal Supply Catalog:

National stock number	Item identification
6515-00-080-0004	FLASHLIGHT, EYE EXAMINING, 1 mm, Fixed Beam, Disposable.
6515-00-185-9482	FLASHLIGHT, EYE EXAMINING, Spot Beam, Disposable.
6515-00-185-9483	FLASHLIGHT, EYE EXAMINING, Slit Beam, Disposable.

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6.5 A suitable filter material is Wratten gelatin filter 47A furnished by Eastman Kodak Co., Rochester, New York 14650.

6.6 A suitable lamp for the type I and type II flashlight is commercial lamp number 222 or 243.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activity:
DoD - MB

Preparing activity:
DLA-DM

Project No. 6515-2095

☆ U.S. GOVERNMENT PRINTING OFFICE: 1977—703-122/1364

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-F-37768 FLASHLIGHT, EYE EXAMINING, DISPOSABLE

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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